

Stop-Time Padding Addendum (unique-only): Max-Mixture Separations and Tail-Sign Deadline Normal Forms

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Positioning inside the overlay body of work

This note records stop-time padding facts that are additive to (and designed to be cited alongside) *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups* [1] and *Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlay Networks* [2]. We therefore do *not* re-derive the kernel-feasible padding model, the Kernel-TV-Pad LP, or the mixture-compiler analysis; those appear in [1]. Likewise, we treat [2] as the canonical reference for random-walk delegation anonymity; the only point of contact here is a modular lemma that lifts stop-time indistinguishability to richer execution views.

Throughout we use the stop-time notation of [1]: a (secret-conditioned) success time $S \in \{1, \dots, S_{\max}\}$ is mapped to an observable stop time $T \in \{1, \dots, T_{\max}\}$ by a single upper-triangular kernel K with $K(s, t) = 0$ for $t < s$. In world $b \in \{0, 1\}$, S has law μ_b , inducing $P_b = \mu_b K$ on T . A TV-privacy budget constrains $\text{TV}(P_0, P_1) \leq \varepsilon$, and the objective is the worst-case mean $M(K) := \max\{\mathbb{E}[T \mid b = 0], \mathbb{E}[T \mid b = 1]\}$.

1 Max-mixture compilers: a baseline strictly richer than timeout-mixtures

The mixture compiler in [1] mixes S with a secret-independent “timeout” draw; here we record a slightly richer, still implementable baseline that snaps a public draw up to feasibility via a max.

Definition 1 (Max-mixture compiler). *Fix $\lambda \in [0, 1]$ and a public distribution Q on $\{1, \dots, T_{\max}\}$. Given $S = s$, output $T = s$ with probability λ ; otherwise sample $U \sim Q$ and output $T = \max\{s, U\}$.*

Timeout-mixtures (as analyzed in [1]) are the special case where Q is supported on $\{S_{\max}, \dots, T_{\max}\}$, so $\max\{s, U\} = U$ always.

Remark 2 (Sanity check at $S_{\max} = 2$). When $S \in \{1, 2\}$, max-mixture compilers already span the global optimum of Kernel-TV-Pad: any feasible kernel can be post-processed by $f(t) = \min\{t, 2\}$ (which cannot increase TV and only decreases the mean) to an optimal kernel supported on $\{1, 2\}$; but every feasible triangular 2×2 kernel has the form $K(2, 2) = 1$ and $K(1, 1) = \lambda$, $K(1, 2) = 1 - \lambda$, i.e., a max-mixture with $Q = \delta_2$.

1.1 Minimal-horizon characterization and a certified strict separation at $S_{\max} = T_{\max} = 3$

At the minimal nontrivial horizon $S_{\max} = T_{\max} = 3$, every feasible kernel can be parameterized as

$$K = \begin{pmatrix} a & b & 1-a-b \\ 0 & c & 1-c \\ 0 & 0 & 1 \end{pmatrix} \quad (a, b, c \in [0, 1], \ b \leq 1-a).$$

Lemma 3 (Polyhedral characterization of max-mixture at $S_{\max} = T_{\max} = 3$). *A feasible kernel K at $S_{\max} = T_{\max} = 3$ is a max-mixture compiler if and only if it satisfies the shared-tail constraint*

$$K(1, 3) = K(2, 3) \quad \Longleftrightarrow \quad c = a + b.$$

Equivalently, the max-mixture family is the intersection of the feasible-kernel polytope with the plane $c = a + b$.

Proof. Under Definition 1, for $s \in \{1, 2\}$ we output 3 exactly when the public draw forces the output to 3, i.e., when $U = 3$ and we are in the $(1 - \lambda)$ branch. Hence $\Pr[T = 3 \mid S = 1] = \Pr[T = 3 \mid S = 2]$, which is $K(1, 3) = K(2, 3)$. In the triangular parameterization this is $1 - a - b = 1 - c$, i.e. $c = a + b$. Conversely, any feasible kernel with $K(1, 3) = K(2, 3)$ can be implemented as a max-mixture: take $\lambda := K(1, 1) = a$ and choose Q so that $(1 - \lambda)Q(3) = K(1, 3)$ and $(1 - \lambda)Q(2) = K(1, 2)$. $\square$

Theorem 4 (Strict separation at the minimal horizon). *Let $\varepsilon = 0.02$ and*

$$\mu_0 = (0.751511627186, 0.194196233939, 0.054292138875), \quad \mu_1 = (0.775286411978, 0.004073598874, 0.220639989)$$

Then the globally optimal kernel-feasible padding value is

$$M^* = 1.592012025884,$$

achieved by the kernel (in the above parameterization)

$$a = 0.841227383338, \quad b = 0.158772616662, \quad c = 0.125049733210,$$

for which $\text{TV}(P_0, P_1) = \varepsilon$. Every max-mixture compiler satisfies the strictly worse bound

$$M_{\text{maxmix}} \geq 2.836458565568.$$

In particular, max-mixture is strictly suboptimal already at $S_{\max} = T_{\max} = 3$.

Remark 5 (Machine-checkable certificate). The instance in Theorem 4 comes with explicit primal and dual witnesses (a tight primal kernel, a feasible dual, and matching objective values) in an accompanying artifact bundle; the stop-time support remains contained in $\{1, 2, 3\}$ even if one enlarges $T_{\max}$, so the separation is genuinely a *minimal-horizon* phenomenon.

2 Tail-sign regime: per-row deadline normal form and a sharp extreme-point corollary

This section records a structural reduction that does not appear in [1]: in a broad sign regime, the TV constraint depends only on the horizon mass, and one may restrict without loss to a low-dimensional “per-row deadline” family.

Let $S_{\max} = T_{\max} = K$ and define the success-time gap $\Delta(s) := \mu_0(s) - \mu_1(s)$.

Definition 6 (Tail-sign condition (TS)). *We say (μ_0, μ_1) satisfies tail-sign if*

$$\Delta(1) \leq 0, \Delta(2) \leq 0, \dots, \Delta(K-1) \leq 0, \quad \Delta(K) \geq 0.$$

Lemma 7 (Sign pattern preservation). *Assume (TS) and let $d := P_0 - P_1 = \Delta K$ be the induced stop-time gap. Then $d(t) \leq 0$ for all $t < K$ and $d(K) \geq 0$. Consequently,*

$$\text{TV}(P_0, P_1) = d(K).$$

Proof sketch. For $t < K$, $d(t) = \sum_{s \leq t} \Delta(s)K(s, t) \leq 0$ because (TS) gives $\Delta(s) \leq 0$ for all $s \leq K-1$. Also $\sum_t d(t) = \sum_s \Delta(s) = 0$, so $d(K) = -\sum_{t < K} d(t) \geq 0$. With this sign pattern, $\|d\|_1 = \sum_{t < K} (-d(t)) + d(K) = 2d(K)$ and hence $\text{TV}(P_0, P_1) = \frac{1}{2}\|d\|_1 = d(K)$. $\square$

Definition 8 (Per-row deadline kernel). *A per-row deadline kernel is specified by parameters $\rho_s \in [0, 1]$ for $s < K$ and sets*

$$K(s, s) = \rho_s, \quad K(s, K) = 1 - \rho_s \quad (s < K), \quad K(K, K) = 1,$$

with all other entries 0.

Theorem 9 (Per-row deadline normal form under (TS)). *Assume (TS). For every feasible upper-triangular kernel K , there exists a per-row deadline kernel $\tilde{K}$ such that*

$$\text{TV}(\mu_0 \tilde{K}, \mu_1 \tilde{K}) = \text{TV}(\mu_0 K, \mu_1 K), \quad \mathbb{E}_{\tilde{K}}[T \mid b] \leq \mathbb{E}_K[T \mid b] \text{ for both } b = 0, 1.$$

In particular, an optimum exists within the family of per-row deadline kernels.

Proof sketch. Define $\tilde{K}$ by preserving each row's ‘‘horizon’’ mass: for $s < K$, set $\tilde{K}(s, K) := K(s, K)$ and put all remaining mass at the earliest feasible time $t = s$ (and $\tilde{K}(K, K) = 1$). Lemma 7 shows that under (TS) the TV constraint depends only on the last column via $d(K) = \sum_s \Delta(s)K(s, K)$, so TV is preserved. Shifting mass from later $t \in \{s, \dots, K-1\}$ to $t = s$ (while keeping $\Pr[T = K \mid S = s]$ fixed) can only decrease the conditional mean, hence $\mathbb{E}_{\tilde{K}}[T \mid b] \leq \mathbb{E}_K[T \mid b]$ for both b . $\square$

Corollary 10 (At most two fractional rows in some optimal extreme point). *Under (TS), there exists an optimal per-row deadline kernel in which at most two of the parameters $\rho_1, \dots, \rho_{K-1}$ lie strictly between 0 and 1.*

Remark 11 (Algorithmic consequence (pointer)). Under Definition 8 and (TS), the optimization reduces to a small LP in (ρ_s) with a single knapsack-type TV budget (Lemma 7). Dualizing yields a one-parameter family of fractional knapsacks, giving an exact finite evaluation method for fixed ε .

3 Dual-shape sparsity beyond generic LP extreme-point sparsity

The Kernel-TV-Pad LP in [1] already implies *generic* sparsity (few nonzeros overall). The following refinement gives a *row-wise* support bound when one has a dual test function with few sign switches.

Lemma 12 (Few-switch duals imply few candidate minimizers). *In the standard TV-linearized dual for Kernel-TV-Pad, suppose the normalized dual test function $\sigma : \{1, \dots, T_{\max}\} \rightarrow [-1, 1]$ is piecewise constant with at most k switch points. Fix a row index s ; the complementary-slackness ‘‘argmin-support’’ rule implies that row s may place mass only on minimizers of an affine score $C_s(t) = tq_s + \Delta(s) \cdot w \sigma(t)$ over $t \geq s$, where $q_s > 0$ and $w \geq 0$ are dual weights. Then every minimizer is attained at the earliest time of one of the constant segments intersecting $\{s, \dots, T_{\max}\}$, so the minimizer set has size at most $k+1$.*

Corollary 13 (Row-support bound). *If an optimal dual witness admits a k -switch normalized test function σ , then there exists an optimal primal kernel whose every row is supported on at most $k+1$ stop times.*

4 Overlay bridge lemma: lifting stop-time privacy to richer execution views

To connect stop-time padding contracts to overlay executions (as in the compiler-calculus view of [1] and the delegation analyses in [2]), it is useful to lift a TV guarantee on T to a TV guarantee on a richer view V .

Lemma 14 (View privacy from stop-time privacy under a failure event). *For each world $b \in \{0, 1\}$ let T_b be a stop time and let V_b be an execution view. Assume there exists an event E_b such that, conditioned on E_b , the view is (up to negligible statistical distance) a post-processing of the stop time:*

$$\text{TV}(V_b \mid E_b, \Phi(T_b)) \leq \text{negl},$$

for some randomized mapping Φ independent of b . Then

$$\text{TV}(V_0, V_1) \leq \mathbb{P}[\neg E_0] + \mathbb{P}[\neg E_1] + \text{TV}(T_0, T_1) + \text{negl}.$$

Remark 15. Lemma 14 isolates a common proof pattern: establish $\text{TV}(T_0, T_1) \leq \varepsilon$ by Kernel-TV-Pad, then identify an event E_b under which the attacker view factors (approximately) through T_b , and separately bound $\Pr[\neg E_b]$.

References

- [1] Anonymous. *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*. Wiki: [[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]. Draft v0.67 (2026-01-21). Published version cited by the same title.
- [2] ChatGPT 5.2-Thinking (extended). *Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlay Networks*. Version 0.56 (single-file, referee-friendly), 2026-01-23.